Fiscal Year:	FY 2018	**Task Last Updated:**	FY 06/04/2025
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Responses of Cardiac Tissue to Simulated Weightlessness		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Ruth.K.Globus@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	04/01/2016	**End Date:**	09/30/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2018 (original end date was 8/31/2017) per A. Ronca/ARC (Ed., 1/9/18)		
Key Personnel Changes/Previous PI:	October 2019: Candice Tahimic has been Co-Investigator.		
COI Name (Institution):	Goukassian, David M.D., Ph.D. (Icahn School of Medicine at Mount Sinai) Schreurs, Ann-Sofie Ph.D. (NASA Ames Research Center / KBRWyle) Shirazi-Fard, Yasaman Ph.D. (NASA Ames Research Center) Tahimic, Candice Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Prolonged bedrest, musculoskeletal disuse, and spaceflight cause complex metabolic adaptations and cardiovascular changes. In addition to exposure to microgravity, the International Space Station (ISS) poses a unique and complex set of physical challenges to physiologic homeostasis; these include confinement, altered gas composition (e.g., elevated ambient CO2), and space radiation. In rodents, short duration spaceflight causes pronounced changes in the expression of proteins and genes related to oxidative metabolism, implicating compensatory responses. The full extent and mechanisms of cardiovascular changes over the long term as a function of age or gender are not understood. We hypothesize that the unique environment of space alters cellular redox balance leading to oxidative damage and ultimately, degenerative changes in cardiac tissue of both males and female rats. This proposed project will test three Specific Aims, taking advantage of the unique tissue sharing opportunity afforded by the University of California Davis experiment using rats, which includes hindlimb unloading, reloading, of both males and females, young and older, using biochemical and gene expression methods. This research will yield new understanding about how the cardiovascular system responds to conditions that simulate weightlessness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this research may facilitate the identification of basic molecular mechanisms that underlie degenerative changes in cardiovascular tissue that arise due to simulated weightlessness. The findings also are relevant to cardiovascular disease on Earth which can result from aging, hormonal changes, immobilization, prolonged inactivity, and bedrest. Identification of specific mechanisms that contribute to the initiation and progression of cardiovascular disease with the hindlimb unloading model may contribute to the development of new countermeasures for spaceflight with potential application to therapies on Earth.
Task Progress:	Spaceflight causes both rapid and delayed effects on the cardiovascular system although the underlying mechanisms are not fully known. Our study aims to determine cardiac changes in response to weightlessness using the rodent hindlimb unloading (HU) model. We hypothesize that long duration exposure to simulated weightlessness and the ensuing recovery leads to changes in cardiac structure and the expression of select genes including those that play a role in protecting against oxidative damage, which all together negatively impact the functional integrity of cardiac tissue. To test this hypothesis, we examined hearts from male (3 and 9 months of age) and female (3 months of age) rats that underwent hindlimb unloading (HU) at various durations up to 90 days and those that underwent reambulation after 90 days of HU. Assays included levels of oxidative damage marker (8-OHdG), gene expression of select antioxidant genes, cardiomyocyte size as assessed by immunohistochemistry, and global transcriptomic analysis (RNAseq). We found that male and female hearts respond differently to HU as indicated by measurements of oxidative damage markers and expression of select antioxidant genes. In older males (9 months old), global transcriptomic analysis (RNAseq) reveals that 14 days of HU led to upregulation of genes involved in the immune response and enrichment of oxidative damage pathways. Compared to age and sex matched controls, changes in cardiomyocyte morphology were observed in older males after 14 days of HU but did not persist at 90 days of HU. In addition, there were no differences between HU groups and control groups after 90 days of re-ambulation post-HU. In older males, long duration HU (90 days) led to persistent downregulation of a key antioxidant gene compared to age and sex matched controls. Taken together, these findings indicate that simulated weightlessness leads to cardiac responses that are age and sex-dependent. Countermeasures for spaceflight-induced cardiac changes may need to be designed based on gender and by age group. Follow-up investigations, beyond the scope of this study, are recommended to determine the impact of HU-induced immune changes on long-term cardiac health.
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Abstracts for Journals and Proceedings	Tahimic CGT, Steczina S, Sebastian S, Terada M, Abegaz M, Hum N,Schreurs AS, Shirazi-Fard Y, Cimini M, Loots GG, Goukassian D, Globus RK. "Simulated weightlessness alters cardiomyocyte structure and transcriptional regulation of mediators related to immunity and cardiovascular disease." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Tahimic CGT, Terada M, Steczina S, Abegaz M, Schreurs AS, Shirazi-Fard Y, Cimini M, Goukassian D, Globus RK. "Effects of simulated weightlessness on the rat heart." Presented at the 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. Abstracts. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Articles in Peer-reviewed Journals	Tahimic CGT, Steczina S, Sebastian A, Hum NR, Abegaz M, Terada M, Cimini M, Goukassian DA, Schreurs A-S, Hoban-Higgins TM, Fuller CA, Loots GG, Globus RK, Shirazi-Fard Y. "Simulated microgravity alters gene regulation linked to immunity and cardiovascular disease." Genes. 2024 Jul 24;15(8):975. https://doi.org/10.3390/genes15080975 ; PMID: 39202335; PMCID: PMC11353732 , Jul-2024
Articles in Peer-reviewed Journals	Tahimic CGT, Globus RK. "Redox signaling and its impact on skeletal and vascular responses to spaceflight." Int J Mol Sci. 2017 Oct 16;18(10): E2153. Review. https://doi.org/10.3390/ijms18102153 ; PubMed PMID: 29035346; PubMed Central PMCID: PMC5666834 , Oct-2017
Articles in Peer-reviewed Journals	Alwood JS, Tran LH, Schreurs AS, Shirazi-Fard Y, Kumar A, Hilton D, Tahimic CGT, Globus RK. "Dose- and ion-dependent effects in the oxidative stress response to space-like radiation exposure in the skeletal system." Int J Mol Sci. 2017 Oct 10;18(10): E2117. https://doi.org/10.3390/ijms18102117 ; PubMed PMID: 28994728; PubMed Central PMCID: PMC5666799 , Oct-2017